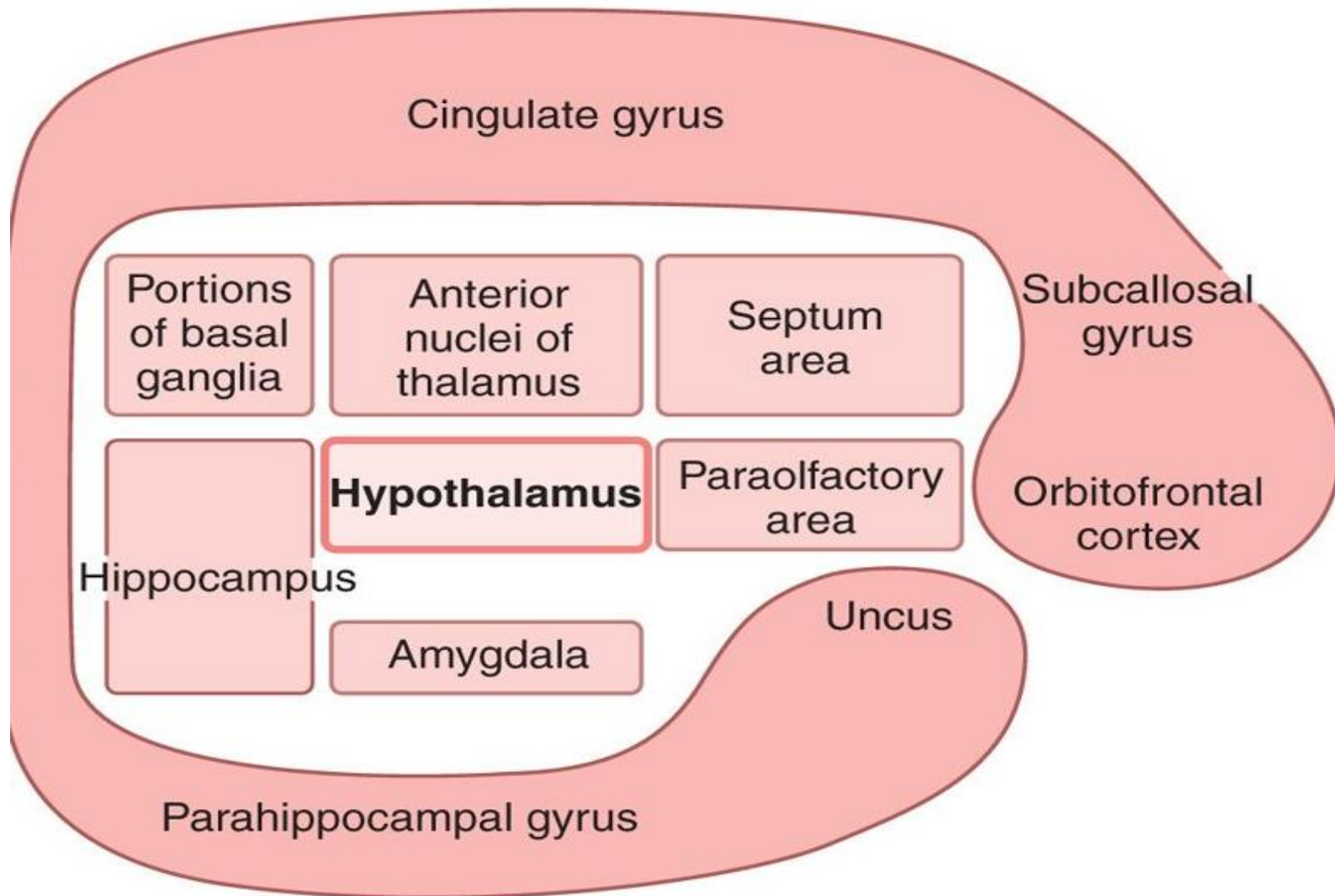

The Limbic System

Components of the limbic system



Limbic System

- Major subcortical structures:
 - **Hippocampus** → learning and memory
 - **Amygdala** → emotions and drives
 - **Septal nuclei** → associated with reward mechanisms
- Highly interconnected and have individual and overlapping functions
- Play major role in memory, emotion, emotional learning and behavior, motivation and reward

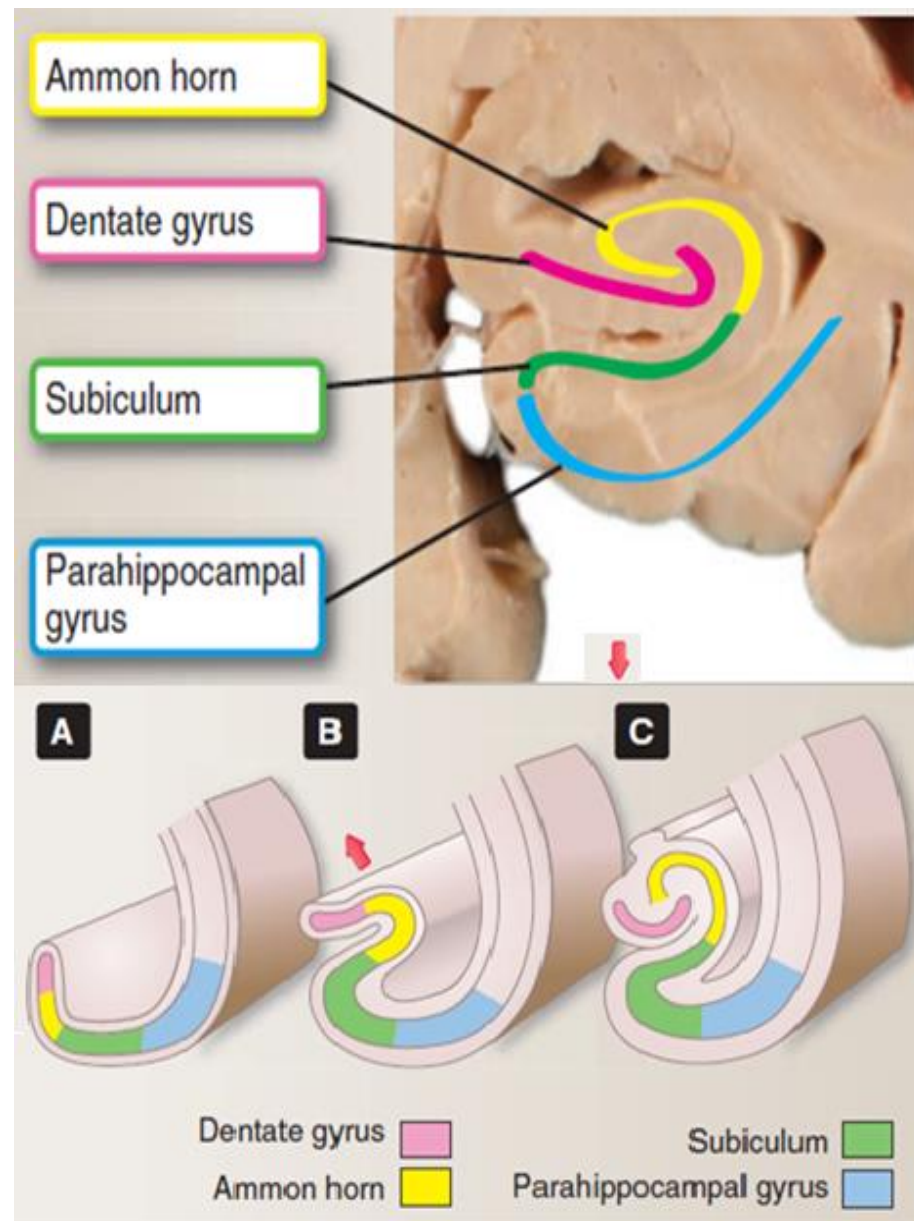
Hippocampal Formation

- A curved sheet of cortex folded into the medial surface of the temporal lobe, occupying the floor of the inferior or temporal horn of the lateral ventricle
- Length $\approx$ 5cm



Hippocampal Formation

- 3 major parts:
- Dentate gyrus
- Hippocampus proper/Ammon horn
- Subiculum
- Parahippocampal gyrus comprises the cortex overlying the hippocampal formation
- Subiculum is a transition zone, continuous with the hippocampus on one side + the parahippocampal gyrus on the other side



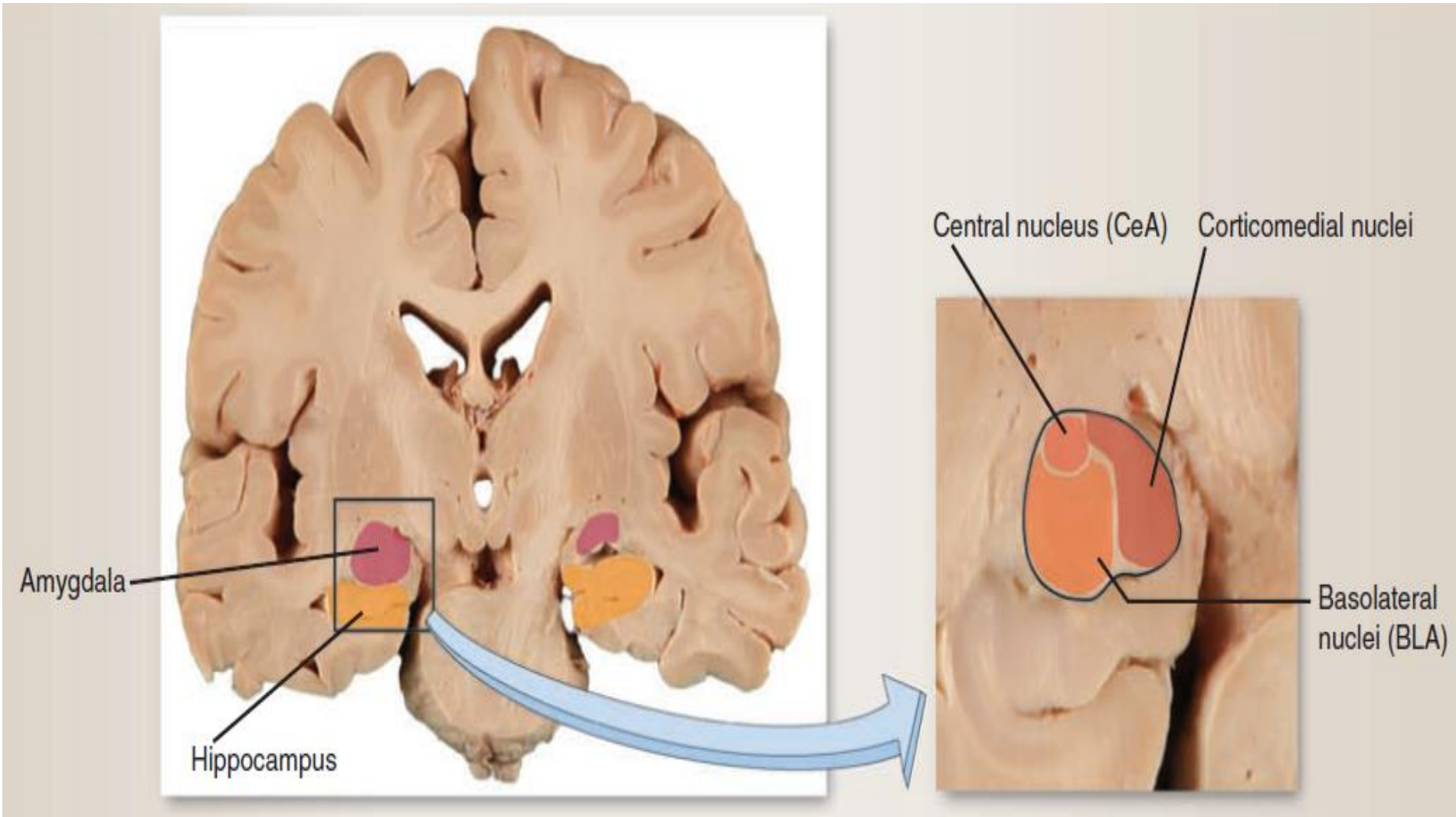
Hippocampus

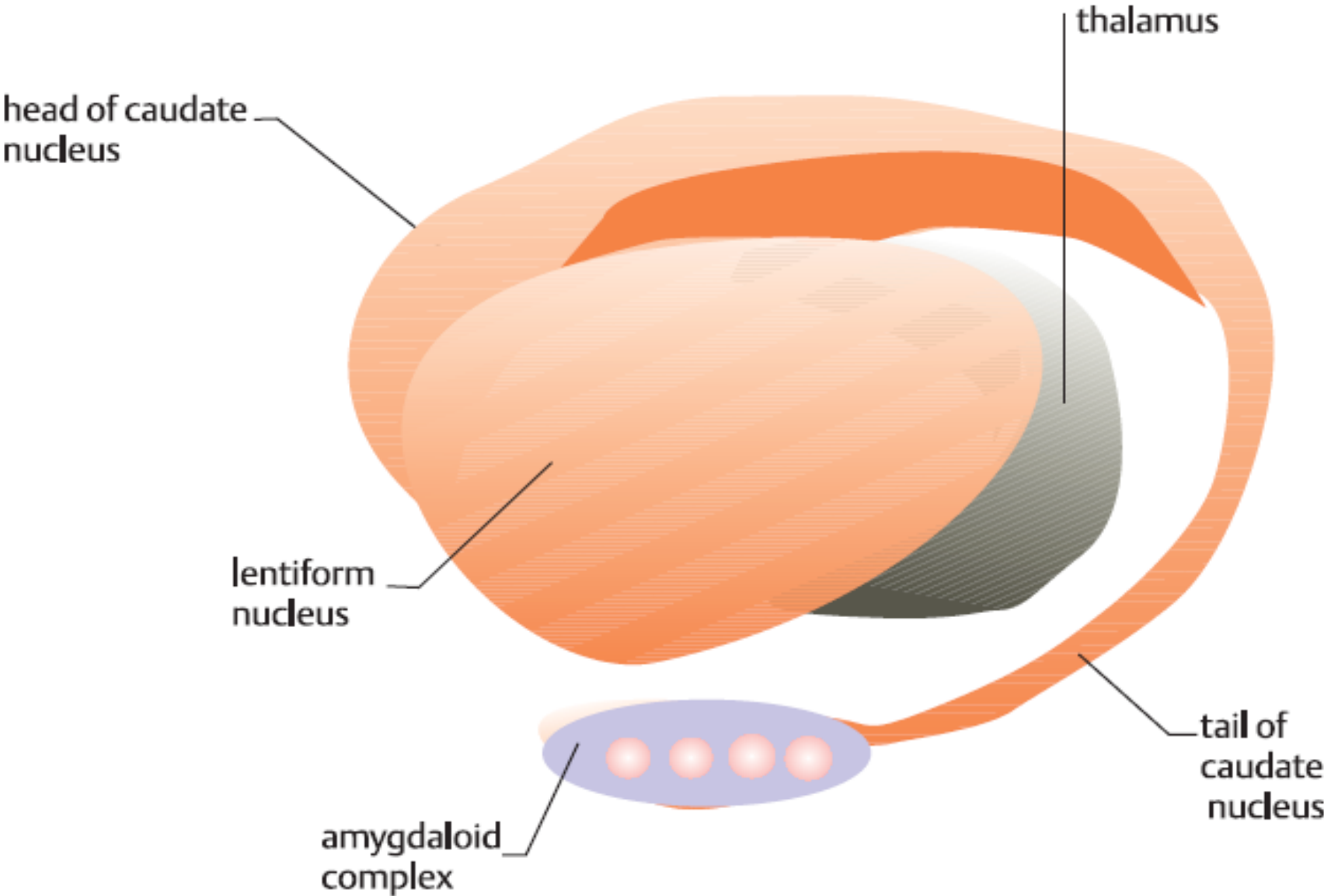
- Functions:
 - Mediation of learning and formation of new memories
 - Expression of emotions and emotional behavior
 - “Spatial navigation” (Human GPS)

Amygdala

- An almond-shaped structure that lies deep in the uncus, just rostral to the hippocampus
- It is buried in the roof of the anterior end of the inferior horn of the lateral ventricle and abuts the tail of the caudate nucleus
- Three morphologically and functionally diversified groups of nuclei:
 1. Basolateral nuclei/BLA
 2. Central nuclei/Central nucleus of amygdala (CeA)
 3. Corticomedial nuclei

Amygdala





Amygdala

1. Basolateral nuclei (BLA)

- The largest group of nuclei and well developed in humans
- Receives information about the **modality and particular characteristics of a stimulus** thru its reciprocal connections with many areas of the cortex:
 - Prefrontal + orbitofrontal + parietal + temporal cortex
 - Cingulate cortex
 - Hippocampus + Parahippocampal gyri
 - Thalamus (ANT, DM) + Striatum

Amygdala

1. Basolateral nuclei (BLA)

- **Efferents** from the BLA project back to the cerebral cortex as well as the thalamus and the central nucleus of the amygdala
- Play a role in **attaching emotional significance to a stimulus**

2. Central nuclei (CeA)

- Reciprocal connections with visceral nuclei of the brainstem + SC
- Also receives input from the BLA
- Important in mediating **general emotional responses**
- Play a role in **regulating visceral responses to emotional stimuli**

Amygdala

3. Corticomedial nuclei

- Receive **olfactory information** from olfactory bulb + thalamus (DM)
- Have reciprocal connections with hypothalamus (ventromedial, **satiety center**) and lateral nucleus (**hunger/feeding center**)
- Provide information about the smell of food to the hypothalamus :
- Play a role in emotional affective responses to food:
 - Pleasant smells stimulate appetite and
 - Unpleasant smells suppress appetite
- Concerned more with **autonomic functions**

Amygdala Connections

- **Afferents:**
 - From thalamus (DM) + widespread cortical areas
 - Sensory input:
 - Visual + Auditory + Somatosensory + Gustatory + Olfactory
 - Inputs from the brainstem:
 - Locus ceruleus + PAG + Raphe nuclei + NTS + RF...

Amygdala Connections

- Locus ceruleus
 - The largest collection of noradrenergic neurons in the pons
 - A center thought to be crucial for attention and cognitive functions
 - Maintains vigilance and responsiveness to unexpected environmental stimuli
- Raphe nuclei
 - Contain cell bodies of serotonergic neurons
 - Serotonin (and possibly NE) is implicated in depression

Amygdala Connections

- PAG (Periaqueductal gray)
 - The gray matter that surrounds the third ventricle and the cerebral aqueduct
 - Produces a profound and selective **analgesia**
 - Stimulation of the PAG matter recruits descending pathways that inhibit nociceptive neurons in the spinal cord

Amygdala Connections

- Nucleus Tractus Solitarius (NTS)
 - Sensory and motor autonomic nuclei (CN IX and X)
 - Mediate two distinctly separate processes:
 1. **GVA functions** → cardiovascular and respiratory reflexes
 - Receive afferents from baroreceptor + chemoreceptor + cardiopulmonary sensory neurons
 2. **SVA functions** → taste
 - Receive taste information from the tongue through SVA neurons

Amygdala Connections

- Reticular Formation
 - Gives rise to three functionally different descending fiber systems
 - These fibers mediate:
 1. Motor functions
 2. Autonomic functions
 3. Pain modulation

Amygdala Connections

- Reticular Formation

1. Motor functions

- Medullary (lateral) reticulospinal tract → powerfully suppresses extensor spinal reflexes
- Pontine (medial) reticulospinal tract → facilitate extensor spinal reflexes

Amygdala Connections

- Reticular Formation

2. Autonomic functions

- Fibers arise largely from the ventrolateral medulla → project to the intermediolateral cell column (IML) of the thoracolumbar cord → excites sympathetic preganglionic neurons in the IML → provide sympathetic innervation to visceral organs

Amygdala Connections

- Reticular Formation

3. Pain modulation

- Enkephalinergic neurons (PAG) → Serotonergic neurons (Raphe nucleus) → Enkephalinergic interneurons (dorsal horn of SC) → primary afferent pain fibers → modulate the activity of pain impulses that ascend in the spinothalamic system

Amygdala Connections

- **Outputs:**
 - Project back to many of the same areas from which the afferents arise
 - Amygdala can send its output directly to cortical and brainstem areas
 - Output to the brainstem may relay thru the hypothalamus → brainstem
- Two major output pathways:
 1. **Stria terminalis**
 2. **Ventral amygdalofugal fibers**

Amygdala Connections

1. Stria terminalis

- A small tract primarily arises from the **Corticomедial nuclei**
- It leaves amygdala, arches over thalamus, running in the groove between caudate nucleus and thalamus → terminates in hypothalamus + ventral striatum + septal nuclei

2. Ventral amygdalofugal fibers

- Arise from **BLA + CeA** → hypothalamus + septal nuclei + ventral striatum and cortex (frontal, prefrontal, cingulate + inferior temporal cortical areas)...

Amygdala

- Functions:
 - With connections to both the hypothalamus and the prefrontal cortex (indirectly), plays a role in both **drive-related behaviors** and the **processing of emotions** that are related to these behaviors:
 - **Emotional response** (anxiety + fear...)
 - **Emotional learning and memory**
 - **Sexual response...**

Amygdala

1. Emotional learning and memory

- An implicit memory function
- Significance of emotional inputs from various cortical areas is assessed primarily by the **basolateral nucleus (BLA)**
- Efferents → hypothalamus → activate the appropriate **visceral and motor responses**
- Via the thalamus (DM) → orbitofrontal cortex (mediates the **conscious perception of emotions**)
- Involved in linking perception with visceral and behavioral responses
+ with memory

Amygdala

1. Emotional learning and memory

- Amygdala works together with memory system of the hippocampus and associated cortical areas (*medial temporal lobe memory system*) during memory formation
- **NB**
 - Events or facts associated with strong emotions (**wedding day, traumatic events...**) are more likely to be remembered than emotionally neutral facts
 - The **link of emotion with the event** solidifies the memory

Amygdala

2. Fear & Fear conditioning

- A form of emotional learning in which a neutral stimulus (*the conditioned stimulus*) becomes associated with an aversive event (*the unconditioned stimulus*)
- Involved in the recognition and interpretation of emotional behaviors or signals from others
- **Amygdala + medial temporal lobe memory systems** are important in mediating the acquisition + storage + expression of fear

Amygdala

2. Fear & Fear conditioning

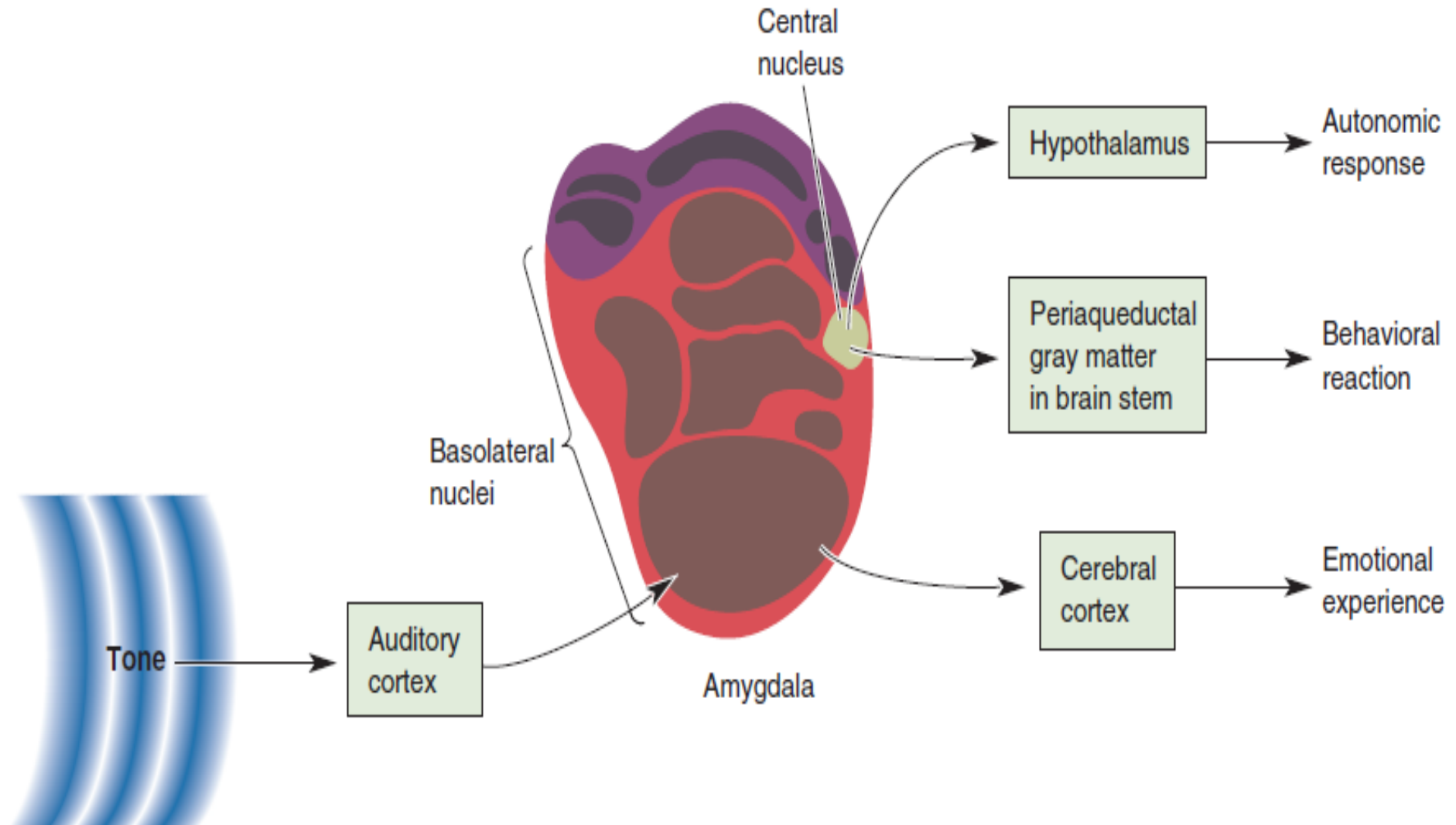
- Neurons in the amygdala can “learn” to respond to stimuli associated with pain
- After such learning, these stimuli evoke a fearful response
- Visceral responses to fear appear to be mediated by the **central nucleus of amygdala (CeA)** thru its connections with the **midbrain, PAG, RF and the hypothalamus**

Amygdala

2. Fear & Fear conditioning: MoA

- Fearful stimuli → BLA → CeA → Efferents project to the:
 - **Hypothalamus** → alter the state of the ANS
 - **PAG** → evoke behavioral reactions via somatic motor system
 - **Cerebral cortex** → emotional experience

Neural circuit for learned fear



Papez Circuit

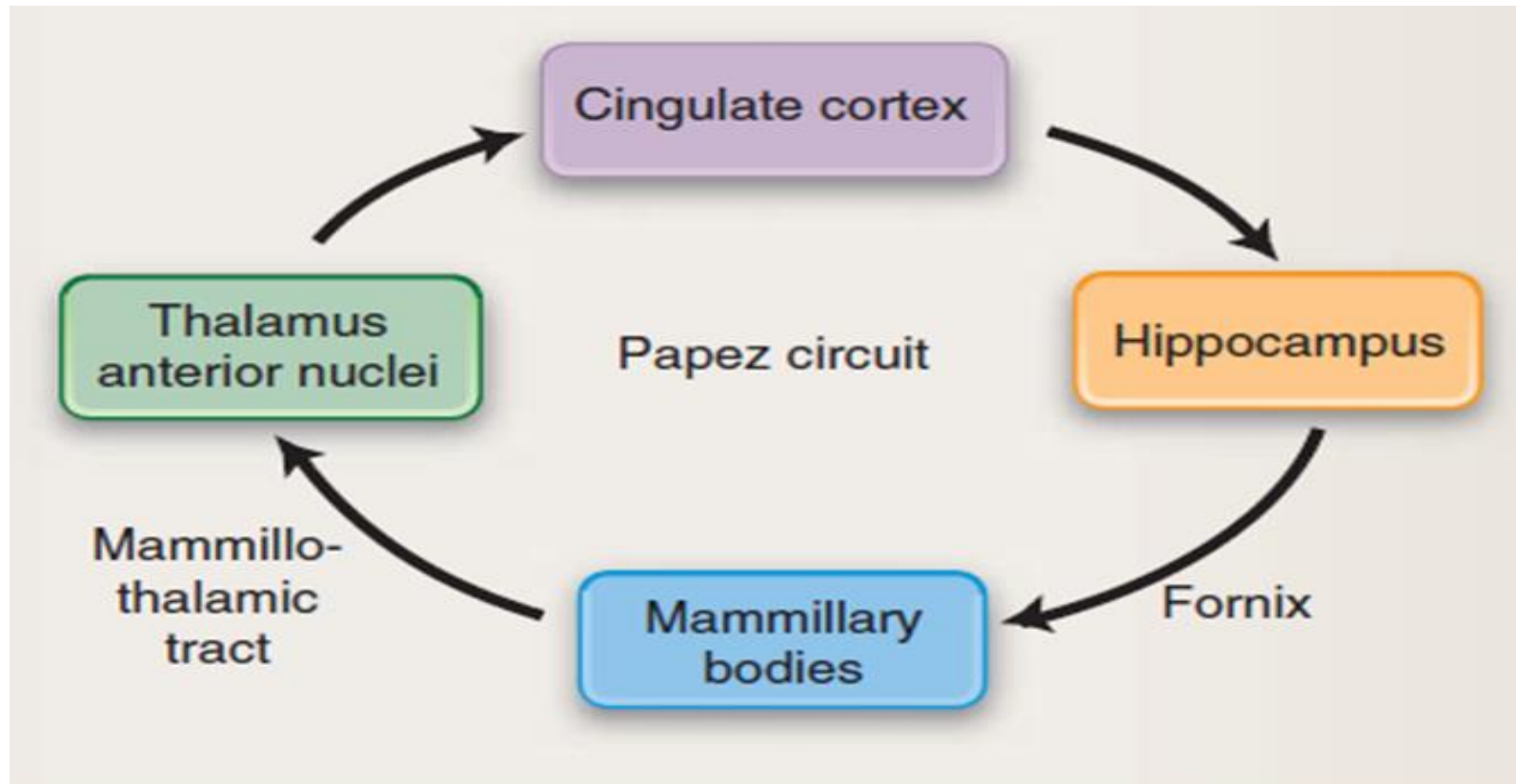
- A major loop circuit of the limbic system.
- Contain a large number of excitatory **reverberating circuits**
- Primarily functions in the cortical control of emotions & memory.
- Its functions are closely associated with hypothalamic functions
- Experience of emotion involves reciprocal interactions between the **diencephalon + cerebral cortex:**
 - ($\because$ emotions reach consciousness + conscious thoughts can affect emotion).

Papez Circuit

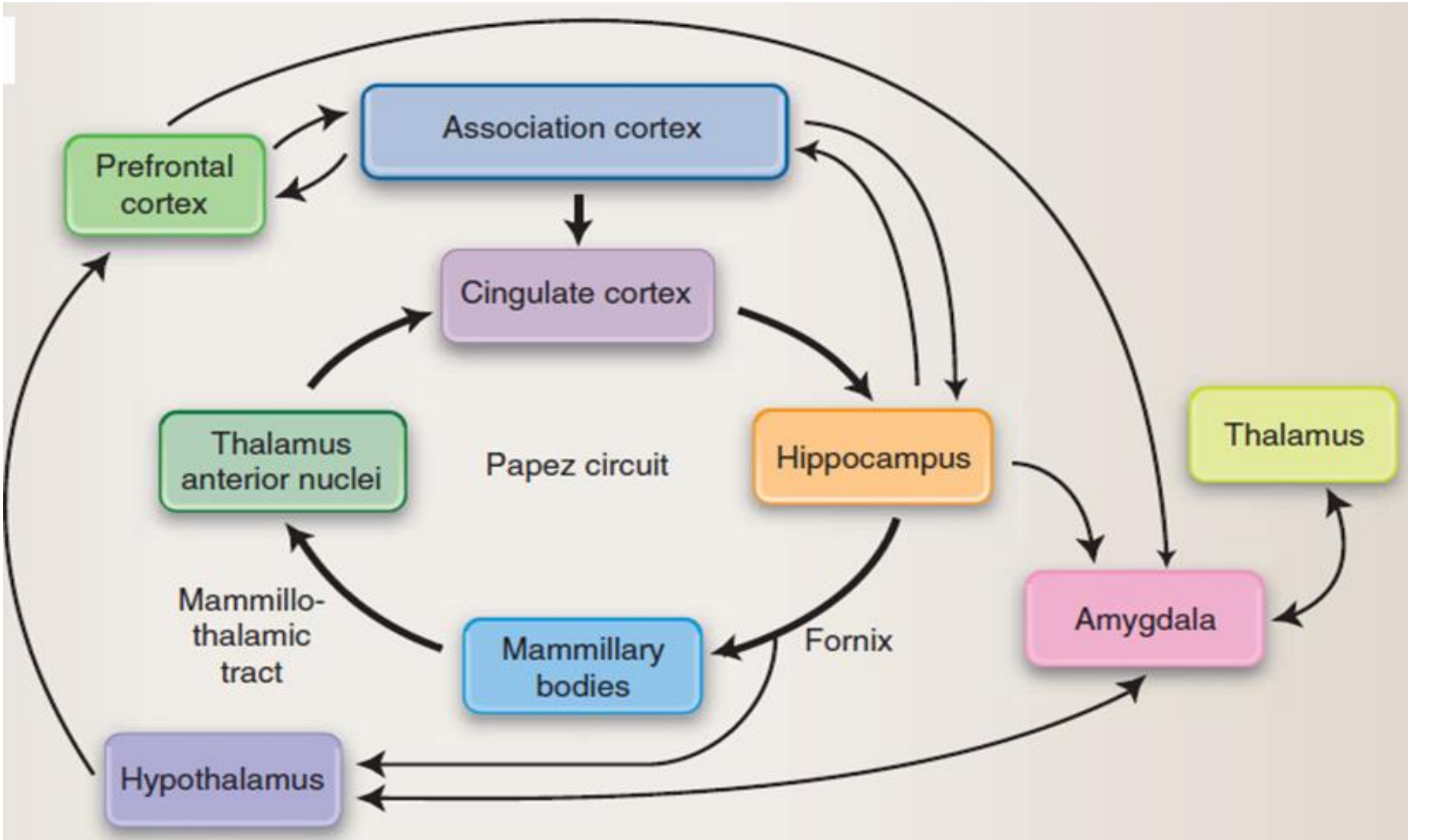
Hippocampus → fornix → mammillary bodies of hypothalamus



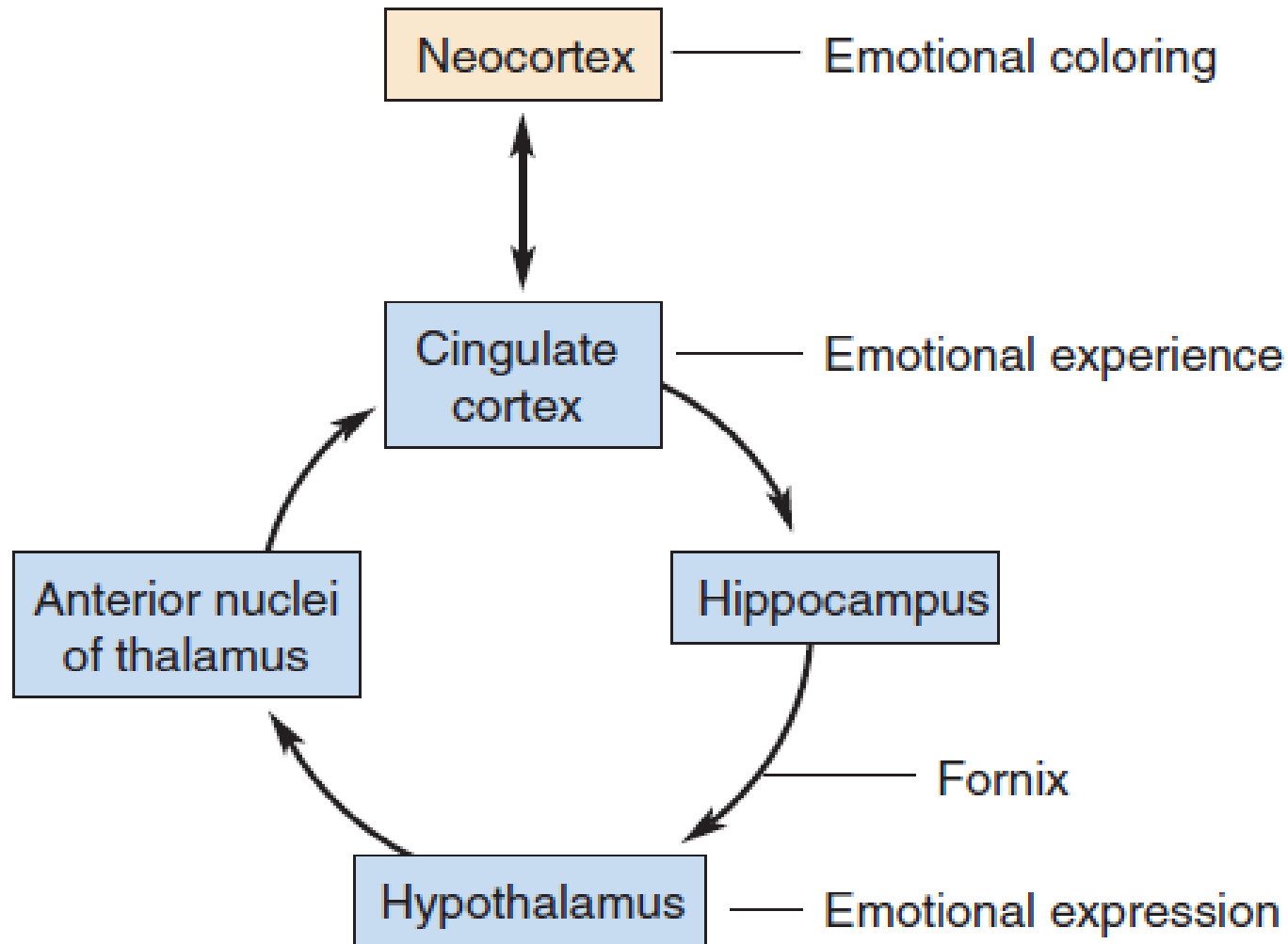
Parahippocampal gyrus ← Cingulate gyrus ← Thalamus (AN)



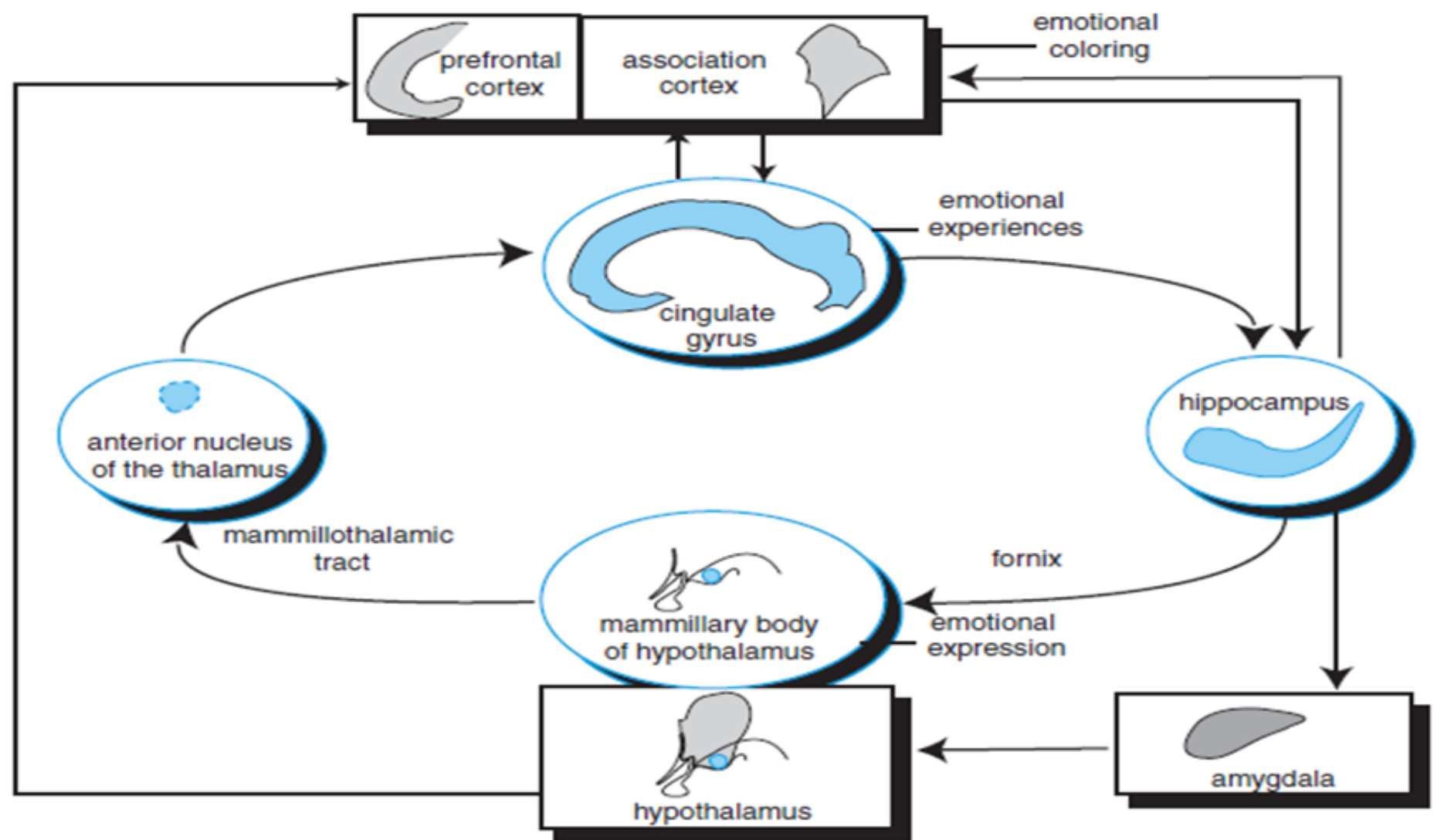
Papez Circuit



Papez Circuit



Papez Circuit



Addiction

Addiction

- Drugs of abuse activate dopaminergic neurons that travel in medial forebrain bundle (like natural rewards and brain stimulation reward)
- All drugs of abuse increase dopaminergic transmission from the VTA to the nucleus accumbens (BUT with chronic intake, the dopaminergic system is impaired):
 - Dopamine receptors are down-regulated → dopamine function is reduced in patients addicted to a variety of different drugs

Addiction

- The result of dopamine receptor down-regulation is that normally rewarding of naturally reinforcing stimuli are less effective (this may, **underlie the negative emotionality seen during withdrawal**)
- Dopaminergic system becomes **sensitized to drugs and drug cues**, resulting in a more intense response to these stimuli
- **Drugs of abuse typically block the dopamine transporter** in the brain reward circuitry, allowing in a large and lasting reward, despite the reduced number of receptors

Dopamine and Saliency in Addiction

- The reason dopamine is so important in addiction is that it **signals saliency**
- In addiction, non drug stimuli take on reduced saliency, whereas stimuli associated with drugs have markedly increased saliency
- **Drug-associated stimuli** activate the **prefrontal cortex** and increase glutamatergic drive to the **nucleus accumbens**
- Increased prefrontal and drug drive results in the greatly increased salience to drugs and drug-associated stimuli, with corresponding increase in **craving and drug-seeking behavior**

Dopamine and Saliency in Addiction

- Altered emotional learning and memory circuitry may also play a role in the increased saliency of drugs and drug cues, and the reduced saliency of to the otherwise pleasurable stimuli (e.g., *food, sex*)
- Association of a drug-induced pleasurable experience with increases in dopamine will result in strong conditioning not only to the drug but also to the stimuli that predict the drug (*e.g., the house of the drug dealer, the neighborhood, syringes...*)
- This could contribute to the enhanced responses to the drug and drug-related stimuli that then overshadow the response to natural rewards

Dopamine and Stress

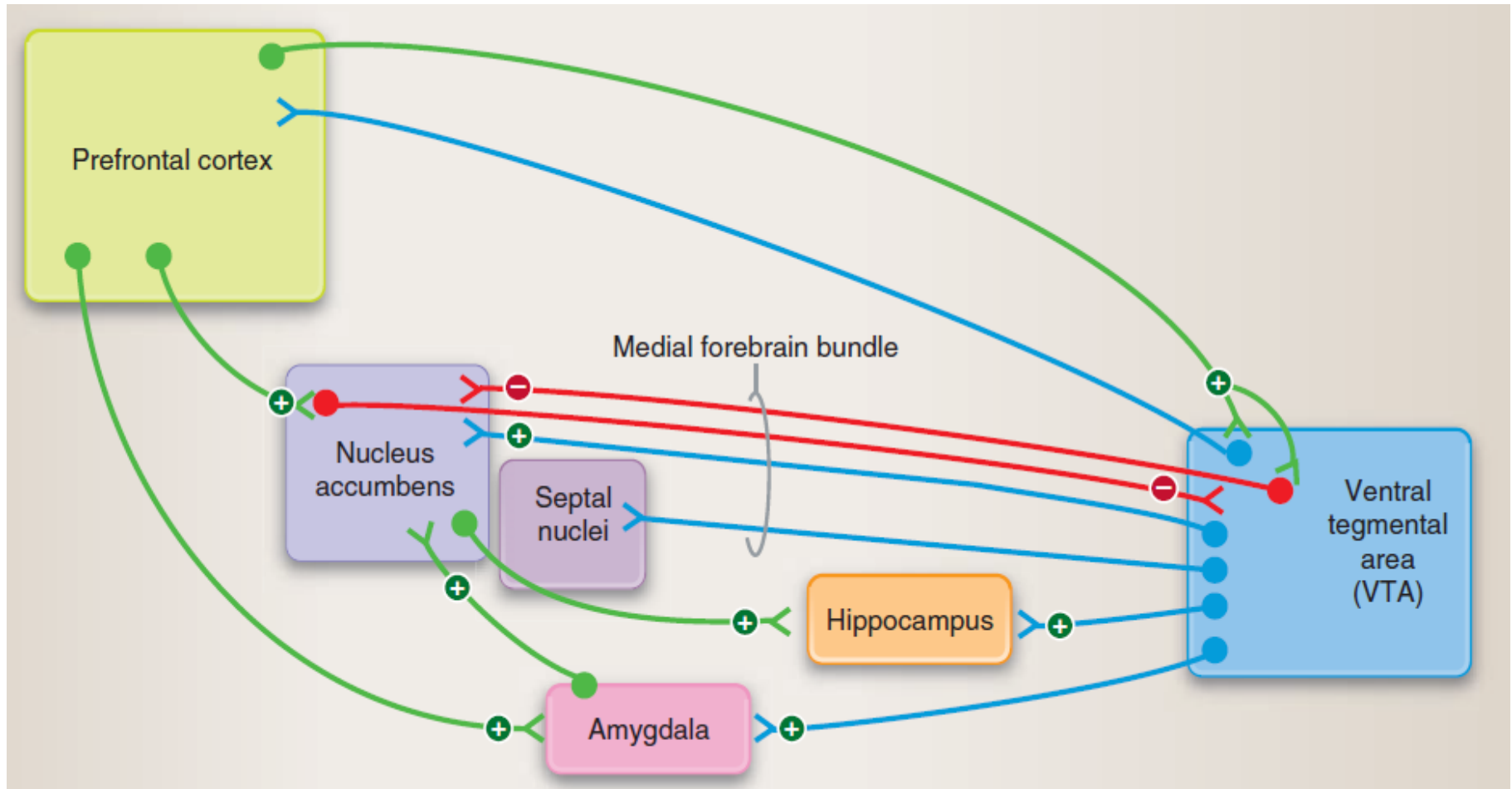
- The dopamine reward circuitry is also responsive to stress
- **Stress and drugs of abuse act similarly** to activate structures involved in the reward pathway (+ including the dopamine neurons in VTA)
- Stress can facilitate the reward associated with initial drug exposure + can increase drug craving and relapse into drug-seeking behavior
- **CRH** in the hypothalamus and in extra-hypothalamic structures (amygdala...) appears to play a role in mediating the effects of stress on the reward circuitry

Reward and Punishment Systems

- The limbic system contains reward + punishment systems
- **Reward system:**
 - Lateral + ventromedial nuclei of the hypothalamus (potent)
 - Amygdala (less potent centers)
 - Thalamus + Tegmentum of the midbrain
 - Stimulation of these systems gives a sense of reward (*i.e., relaxation, pleasure + satisfaction*)
 - The subject is motivated to approach or repeat the rewarding experience (*the approach reaction*)

Reward circuitry

- It is processing positive affect as it is for negative affect (fear conditioning)



Reward and Punishment Systems

- **Punishment system:**

- Periventricular nucleus of the hypothalamus + thalamus
- PAG + Amygdala + hippocampus
- Stimulation of the punishment system gives a sense of punishment
(i.e., displeasure, fear and terror)
 - The subject is motivated to avert the punishing experience *(the aversion reaction)*

Memory

Memory

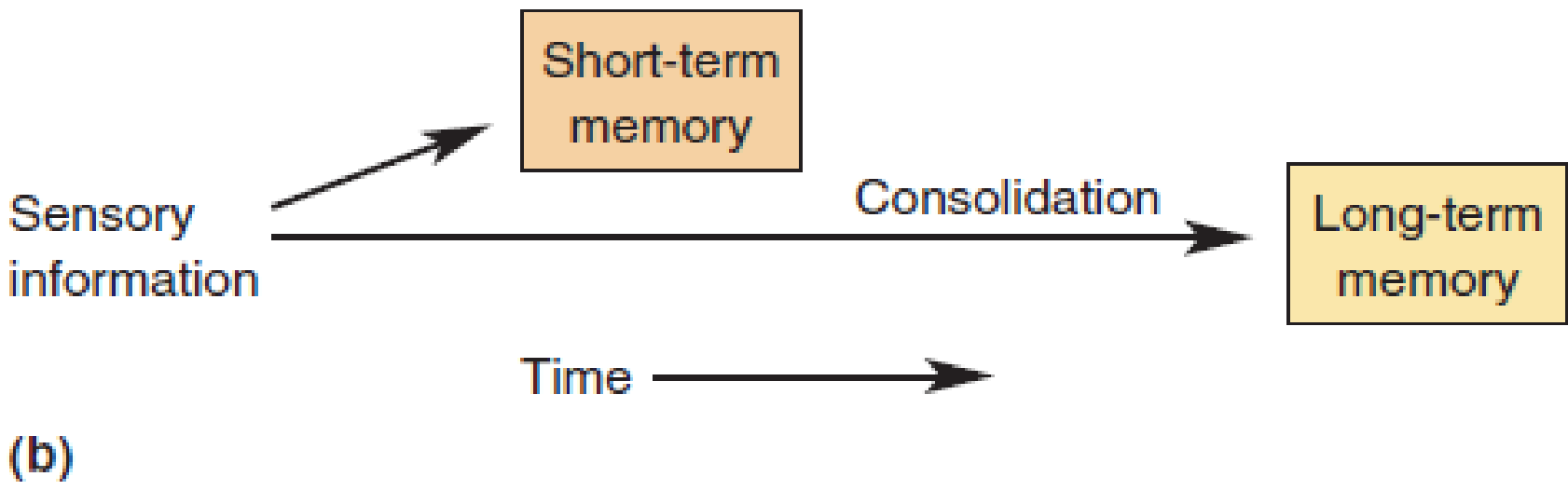
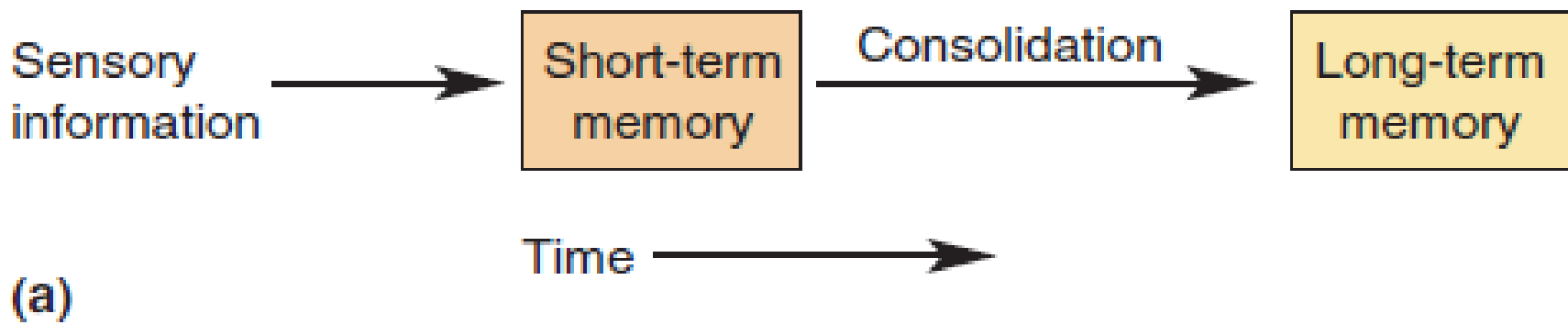
1. Short-term memory/working memory

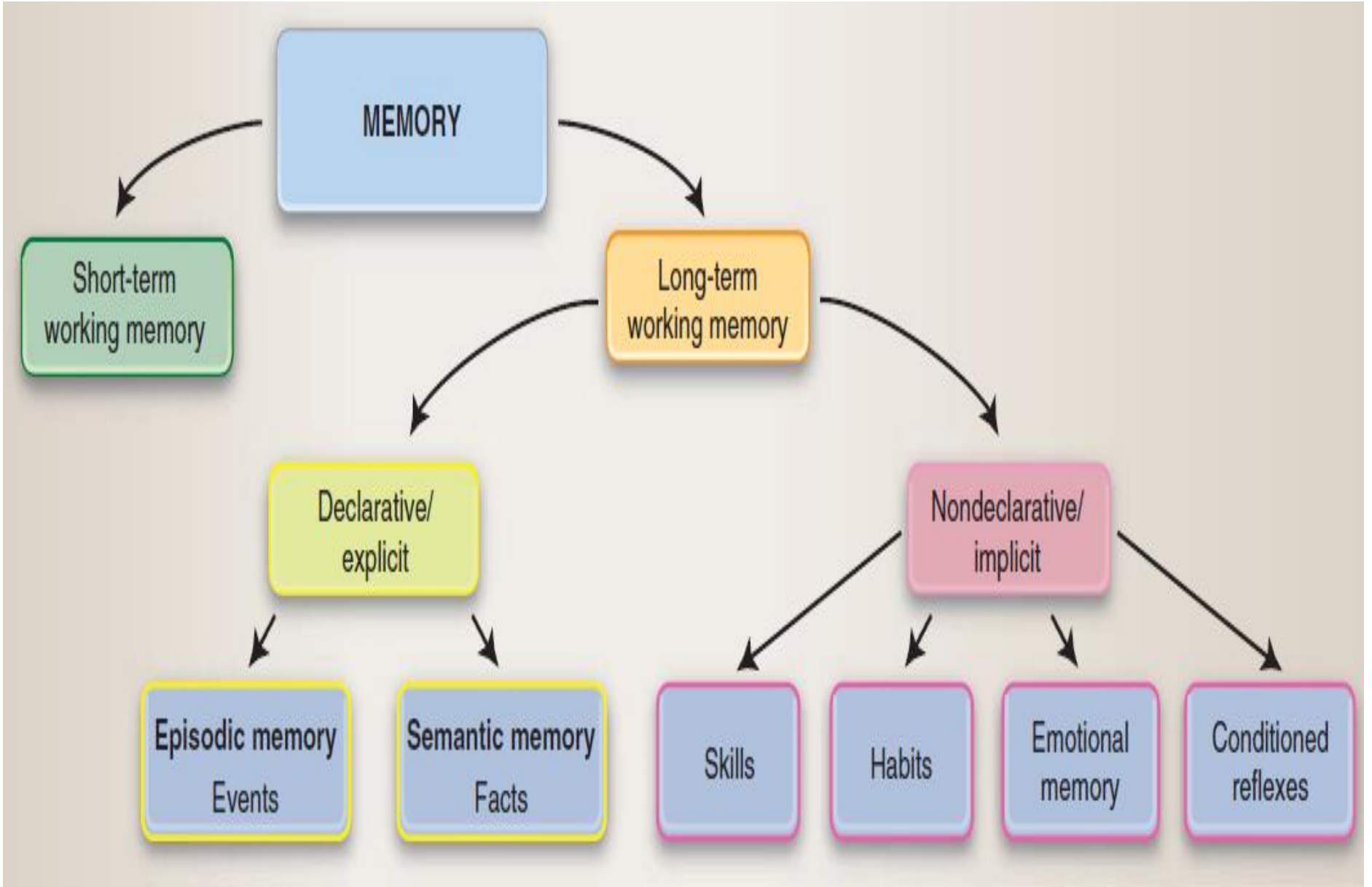
- Involves the short-term maintenance of information in memory and sometimes the manipulation of that information to achieve an immediate goal
- Depends primarily on the Prefrontal cortex

2. Long-term memory

- Explicit/Declarative memory
- Implicit/Non-declarative memory

Memory





Declarative Memory

- Involves memories of events or facts that are **accessible to consciousness** and can be expressed explicitly (*expressing all details in a clear and obvious way*)
- **Two types:**
 1. Episodic memory → Events
 2. Semantic memory → Facts

Declarative memory
(Medial temporal lobe; diencephalon)

Facts

Events



Declarative Memory

1. Episodic memory:

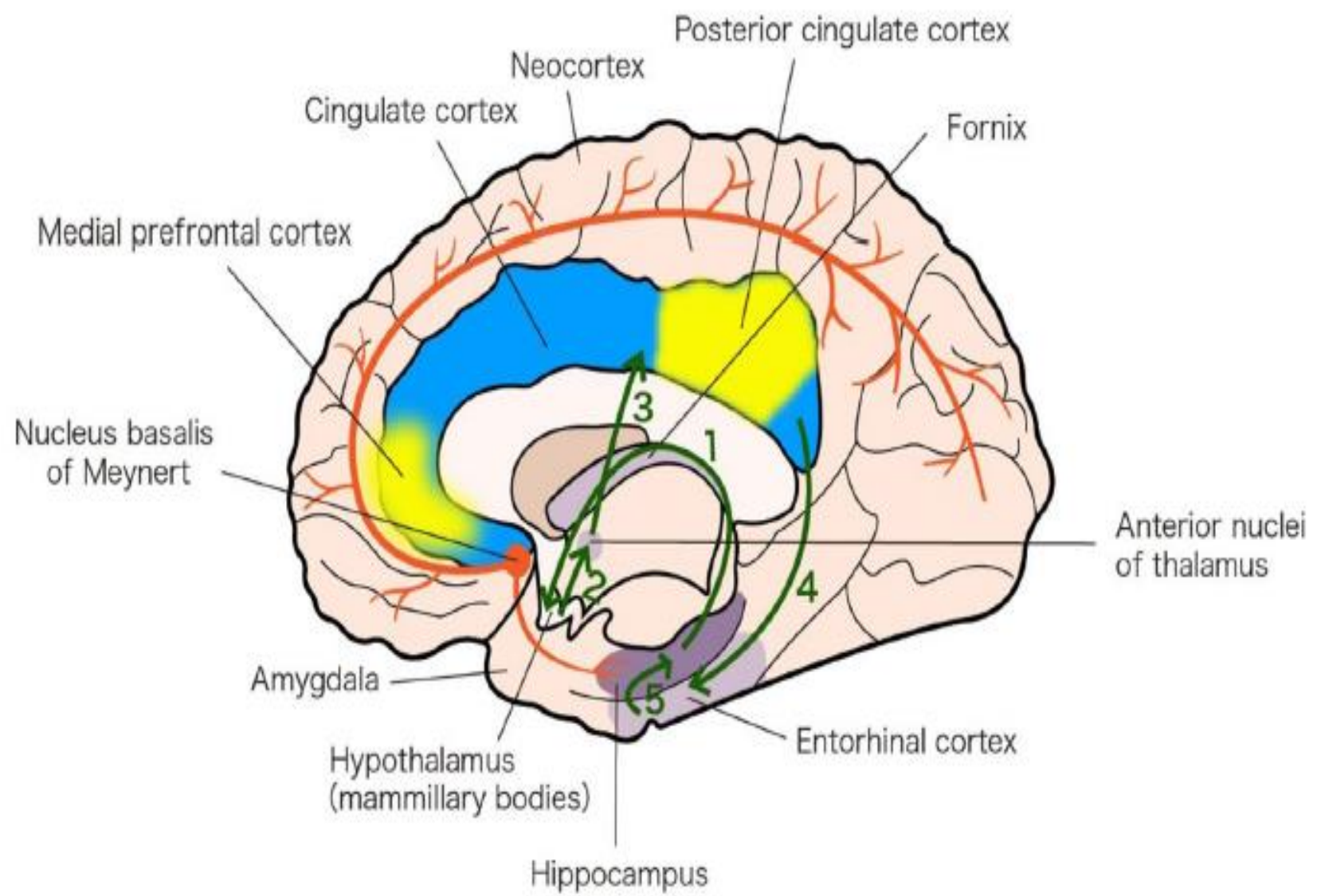
- Involves memory of **events**
- Includes the ability to learn + store + retrieve information about experiences that occur in everyday life (**time + place and details** about the event itself)

2. Semantic memory:

- Involves the **knowledge of facts** that have been learned, but of which the source of the original information is typically not known...

Declarative Memory

- **NB**
 - Explicit memory requires the **Medial temporal lobe memory systems** → hippocampus + associated cortical areas (Entorhinal cortex + parahippocampal gyrus) + widespread neocortical association areas and the pathways that connect these structures with other parts of the brain



Memory circuits

Nondeclarative Memory

- Involves memories that manifest as **subconscious behavioral or physiological responses** to events or stimuli
- Includes several forms of learning that occur during performance of a task:
 - Skills + habits
 - Emotional memory
 - Conditioned reflexes

Nondeclarative Memory

1. Skills and habits

- Like driving, swimming, riding a bicycle...
- Memory of skills and habits depends on the **striatum** (caudate nucleus and putamen), **motor areas of the cortex**, and the **cerebellum**

2. Emotional memory (Emotional associations)

- Involves a change in behavior toward a previously neutral stimulus as a result of experience and depends on the **Amygdala**

3. Conditioned reflexes

- Primarily depends on **cerebellum** (eg., Pavlov's dog)

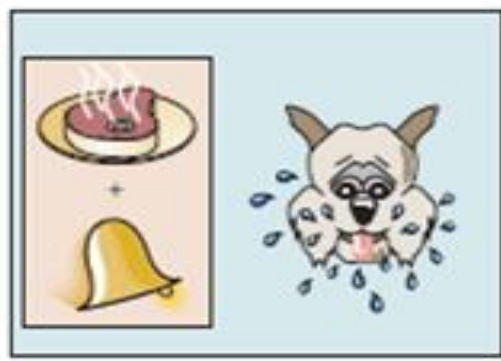
Nondeclarative memory

Classical conditioning

Procedural memory:
skills and habits
(Striatum)

Skeletal musculature
(Cerebellum)

Emotional responses
(Amygdala)



Amnesia

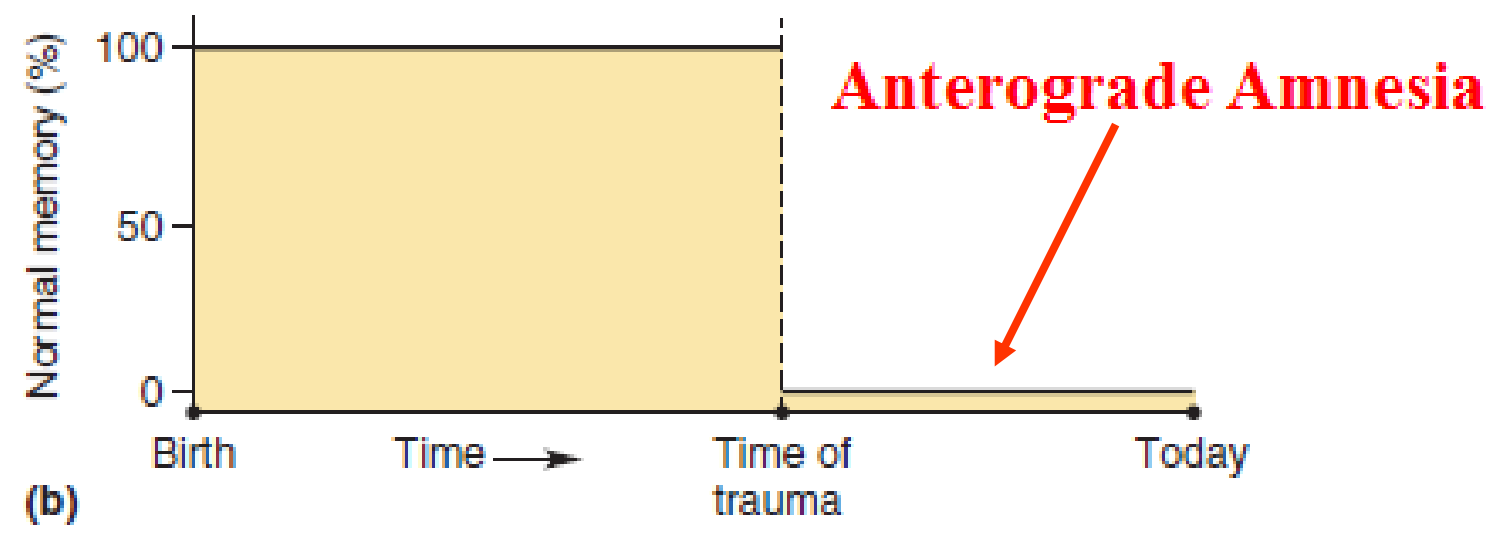
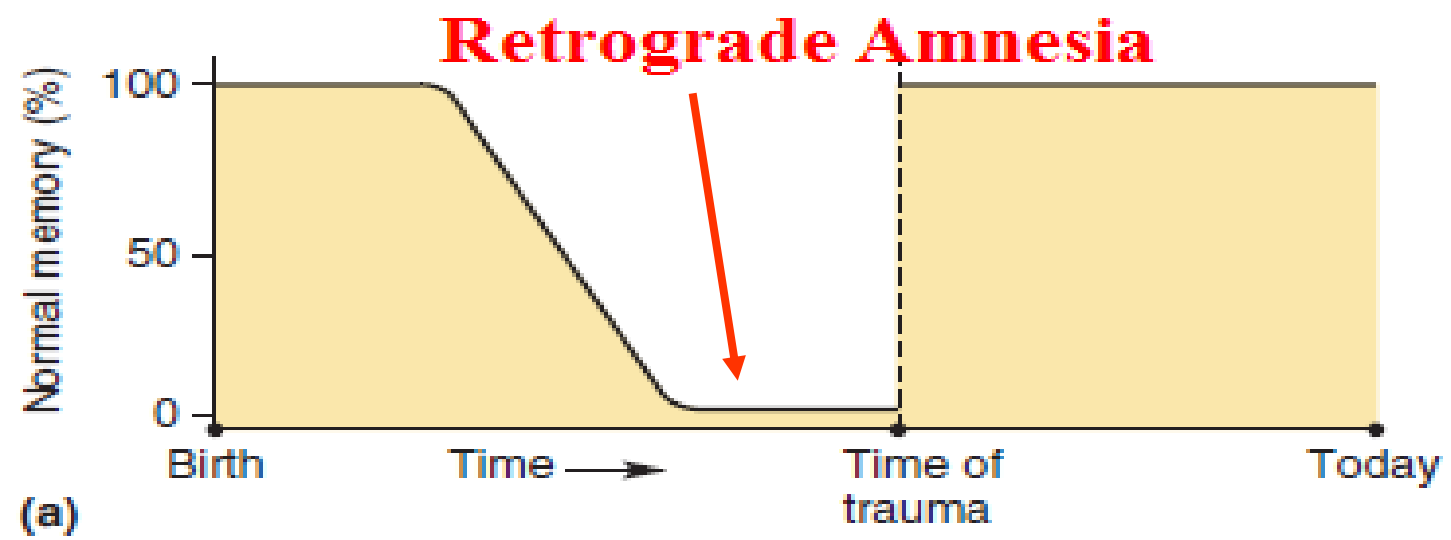
- Loss of memory and/or the ability to learn
- Concussion, chronic alcoholism, encephalitis, brain tumor, and stroke can all disrupt memory
 - **Dissociated amnesia**
 - If amnesia is not accompanied by any other cognitive deficit
 - i.e., memory problems are dissociated from any other problems
 - **Absolute amnesia** → Absolute memory loss about past events
- Memory loss can be manifested in two different ways:
 - **Retrograde amnesia**
 - **Anterograde amnesia**

Retrograde Amnesia

- Events of a period of time before the trauma are forgotten, but memories from the distant past and the period following the trauma are intact
- In severe cases, there might be complete amnesia for all declarative information learned before the trauma
- More often, follows a pattern in which events of the months or years preceding the trauma are forgotten, but memory is increasingly strong for older memories
- **N.B**
 - The **graded loss of old memories** across time apparently reflects the changing nature of memory storage

Anterograde Amnesia

- Inability to form new memories following brain trauma
 - Events before the trauma can be remembered, but there are no memories for the period following the trauma
- In mild cases, learning may be slower + require more repetition than normal
- In severe cases, the person might be completely incapable of learning anything new
- **NB**
 - In clinical cases, there is often a mixture of retrograde and anterograde amnesias of different degrees of severity



Anterograde Amnesia

- **Transient Global Amnesia:**

- A form of amnesia that involves a much shorter period of time
- Sudden onset of anterograde amnesia lasts only for a period of minutes to days, often accompanied by retrograde amnesia for recent events preceding the attack
 - During the spell, the person may appear disoriented and ask the same questions repeatedly, but conscious, and measures of working memory are normal
 - In a matter of hours, the attack usually subsides, and the person is left with a permanent memory gap